

1 Light-sharing control for tracker-based agrivoltaics
2 under a yield floor: what learning adds and what
3 shade-response uncertainty costs

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6 **Abstract**

Single-axis trackers let agrivoltaic arrays shift the daily rotation from maximum electricity towards letting light through (light sharing). We quantify its value under a yield-retention floor of 90% of open-field yield with a differentiable simulator of the whole system. The crop response to shade is the least certain part: a hierarchical model fitted to 428 observations from 58 field studies predicts a new study of the four staple crop types only to about 0.22 in relative yield, no better than proportional loss, so we turned the learned curves and four field trials into ensembles of plausible responses by gradient descent through the simulator. On the 92 held-out crop–site pairs of the main split, light-sharing rules allowed arrays 2.1–2.9 times as dense as static arrays at equal mean retention and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24 (1.27 for a neural controller), with a phenology rule needing no forecast the best; the floor was met in 69–76% of seasons. Pooled over three splits (76 pairs), the neural controller exceeded the written rule by 0.044 LER and a rule tuned on the same seasons by 0.018 (95% interval 0.012–0.024); a CMA-ES-optimised schedule was 0.013 behind it, and training on ensembles or a farm-measurable biomass signal did not help. Design mattered more: a density chosen under the mean response left about 70% of pairs at the floor when another plausible response held (oracle 85–87%); a floor raised by 0.02 (85–88% of pairs, 0.06–0.07 LER below the mean-response design) was as good as or slightly better than an ensemble-quantile rule, and none of these rules was safe outside the ensemble (7–33% of pairs under three fixed responses). Under illustrative costs, light sharing breaks even with the open field at 70–92 USD MWh^{−1} and static arrays at 59, and a conventional plant is more profitable above 48. For maize and soybean (53% of the upscaled

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14 PWh yr⁻¹) the bias of the simulator is uncertain.

7 *Keywords:* Agrivoltaics, Light sharing, Solar tracking, Food–energy nexus,
8 Crop modelling, Differentiable simulation, Hierarchical shrinkage model

9 1. Introduction

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017; IPCC, 2021), and a new generation
17 of climate and crop models shows these impacts emerging earlier than pre-
18 viously expected (Jägermeyr et al., 2021). Croplands are also where solar
19 power potential is greatest (Adeh et al., 2019). They are flat, sunny, acces-
20 sible and close to rural electricity demand, so converting farmland into solar
21 farms is tempting and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
29 climate. It lowers canopy temperature and evaporative demand, conserves
30 soil moisture and improves water-use efficiency, especially in drylands (Mar-
31 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
32 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a can-
33 didate adaptation measure against heat and drought stress, as reviewed by
34 Mamun et al. (2022), in addition to a mitigation measure through renew-
35 able electricity. The same shade, however, also reduces the light available for
36 photosynthesis. Regulators increasingly require AV systems to retain most
37 of the reference crop yield. The German pre-standard DIN SPEC 91434, for
38 example, requires at least 66% of the reference yield (DIN, 2021), and other
39 schemes and many farmers set stricter floors. How much electricity can be
40 produced without sacrificing food security depends on the crop, the local

41 climate, the year and the growth stage.

42 Most AV designs fix the geometry of the array, i.e. the ground coverage
43 ratio (GCR, module area per ground area) and the tilt or tracking rule, from
44 generic rules of thumb. Single-axis trackers add a degree of freedom: the
45 modules can be rotated away from the sun to let light through when the crop
46 needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the
47 sun to maximise electricity. Choosing the rotation every day is a sequential
48 decision problem. The value of light depends on the crop’s phenological stage,
49 soil water and upcoming weather, and the effect of a decision appears in the
50 harvest months later. Recent work in this journal has used evolutionary
51 multi-objective optimisation to lay out PV arrays on farmland (Zhang et al.,
52 2026). Machine learning for crop management has mainly used reinforcement
53 learning with black-box simulators (Gautron et al., 2022), which is sample-
54 inefficient and hard to scale to global, multi-decadal assessments.

55 Two methodological strands are relevant. Differentiable process models,
56 which couple physical simulators with learning, allow parameters and poli-
57 cies to be fitted by gradient descent through the simulator (Reichstein et al.,
58 2019; Shen et al., 2023), and we use them for both calibration and control.
59 Decision making under deep uncertainty asks for designs that perform accept-
60 ably across plausible models and not only under the best estimate (Marchau
61 et al., 2019), and scenario-based designs have finite-sample guarantees un-
62 der convexity assumptions that we do not meet (Calafiore and Campi, 2006;
63 Campi and Garatti, 2008); our design rules are heuristic counterparts and
64 we compare them with a simple safety margin.

65 Two questions follow. First, how much of the electricity that an agri-
66 voltaic system can produce under a food-security floor comes from the array
67 density, and how much from dynamic light-sharing control of the trackers?
68 Second, how robust is that answer to the weakest part of any such study, the
69 crop response to shade, and does a controller learned from a simulator add
70 anything to a well-designed hand-written rule? Answering them requires
71 a simulator that couples the physics of light, PV, microclimate and crop
72 growth, a protocol in which all controllers face the same design constraints,
73 a check of the shade response against field data, and enough sites and seasons
74 to separate real effects from noise.

75 This paper makes three contributions. The first is a quantification of
76 light-sharing control under a yield floor. Light sharing is parameterised by
77 one daily level between full backtracking and the rotation that minimises the
78 shadow on the crop, and we compare static arrays, a seasonal schedule, a rule

that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three training seeds and three spatial splits. The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS. The crop response to shade is learned from the open data of a meta-analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023), and propagated through the simulator as an ensemble.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever. At the same mean yield retention of 90%, arrays with light-sharing control carry 2.1 to 2.9 times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The crop response to shade can be learned from data. A hierarchical model fitted to 428 observations from 58 field studies predicts a held-out study of the four staple crop types to about 0.22 in relative yield, no better than proportional loss (it is better only for crop types the simulator does not represent), so between-study spread is large. The simulator is close to these curves for cereals and potato but milder than the meta-analysis for maize and soybean, whereas one maize trial is milder still; the gains for those two crops are therefore uncertain and

- possibly overestimated (Section 3.7).
4. A learned controller adds a small gain over the best rule: 0.044 LER over the written phenology rule and 0.018 over a rule tuned on the same training seasons. Training it on ensembles of shade responses, or giving it a farm-measurable biomass signal, does not help, and with the wider ensembles it did not retain its margin over the written rule in our training budget (Section 3.8).
 5. The bottleneck is the uncertainty of the crop response that enters the design. A density chosen under the mean response leaves about 70% of crop–site pairs at the floor when another plausible response holds (oracle 85–87%). A rule built on the quantiles of an ensemble of responses raises this by 9 points for 0.04 LER, but a floor raised by 0.02 under the mean response is as good as or slightly better at equal compliance, and no rule is safe outside the ensemble, so the ensemble is useful to size a margin and not to replace it (Section 3.8).
 6. Under illustrative costs, light sharing breaks even with the open field at 73 to 92 USD MWh^{−1}, learning is worth about 2 kUSD ha^{−1} yr^{−1} over a tuned rule at 90 USD MWh^{−1}, and a conventional plant replacing the crop is more profitable at every price (Section 3.9).
 7. At the yield floor, agrivoltaics on 5% of the world’s staple cropland could supply about 14 PWh yr^{−1}, but only with about 10 TWp of modules, several times the installed PV capacity of the world (Section 3.11).

2. Materials and methods

2.1. Global cropland sites, weather and crop calendars

We studied five staple crops that together cover about 670 Mha of harvested area and 580 Mha of physical cropland: wheat, rice, maize, soybean and potato. Harvested areas (rainfed and irrigated) were taken from the SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1° cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn without replacement with probability proportional to harvested area. An unweighted average over the sample therefore approximates an area-weighted average over global production. The 400 crop–site pairs lie in 381 distinct cells in 50 countries (Fig. 2). Sowing and maturity dates came from the GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat we used the winter-wheat calendar unless the coldest-month mean temperature was below −8 °C, in which case the spring-wheat calendar was used.

152 Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks,
 153 2018): global and diffuse shortwave irradiance, maximum, minimum and
 154 dew-point temperature, precipitation and wind speed at 2 m. Each site con-
 155 tributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days
 156 after sowing, which gives 7600 site-seasons.

157 Climate change was represented by the global warming levels +1.5, +2
 158 and +3 °C relative to 2001–2020. Daily maximum, minimum and dew-point
 159 temperatures were shifted uniformly, which keeps relative humidity approx-
 160 imately constant and raises the vapour-pressure deficit. The atmospheric
 161 CO₂ concentration was set to 390, 450, 500 and 600 ppm for the baseline and
 162 the three warming levels. Precipitation and irradiance were left unchanged.
 163 This delta approach isolates the thermal signal of warming; its limitations
 164 are discussed in Section 4.5.

165 2.2. A differentiable agrivoltaic simulator

166 The simulator advances all site-seasons in parallel with a daily time step
 167 for the crop and an hourly time step for radiation and PV (Fig. 1). All equa-
 168 tions are implemented as PyTorch tensor operations (Paszke et al., 2019), so
 169 that yields and electricity are differentiable with respect to the design and
 170 control variables.

171 *Hourly irradiance.* Daily global and diffuse irradiance were disaggregated
 172 into hourly values with the Collares-Pereira and Rabl (1979) and Liu and
 173 Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance
 174 on the horizontal is the difference between the two. Solar position follows
 175 the standard declination and hour-angle equations.

176 *Array geometry and PV electricity.* We consider elevated horizontal single-
 177 axis trackers with north–south axes. The axis height is 1.25 module widths,
 178 e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The ro-
 179 tation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width
 180 divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this
 181 keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the
 182 rows when the modules are horizontal, which is the stow position during field
 183 operations. Let ψ be the solar zenith angle projected onto the east–west ver-
 184 tical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,
 185 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ oth-
 186 erwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on

187 to the sun, which minimises the shadow on the crop. Given a light-sharing
 188 level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

189 Plane-of-array irradiance combines three terms: beam irradiance on the mod-
 190 ule, reduced by the fraction of the module shaded by the neighbouring row,
 191 $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected
 192 light (albedo 0.2). The cell temperature follows the NOCT model (45 °C),
 193 with an hourly air temperature interpolated sinusoidally between the daily
 194 extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with
 195 $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature
 196 losses and system losses. The installed capacity is $2.1 g \text{ MWp ha}^{-1}$. A con-
 197 ventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the
 198 energy reference E_{PV} .

199 *Validation of the PV model.* We compared the simulated specific yield of an
 200 isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2
 201 (European Commission, Joint Research Centre, 2022), for a horizontal north-
 202 south single-axis tracker with 14% losses, at 30 randomly chosen sites (29
 203 returned data). Across sites, the simulated and PVGIS specific yields were
 204 correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean
 205 absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and
 206 with the different irradiance databases (NASA POWER vs. SARA H2, ERA5
 207 and NSRDB). Electricity estimates are therefore, if anything, conservative.

208 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction
 209 of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The
 210 fraction of sky diffuse reaching the ground is a two-dimensional isotropic
 211 sky-view factor. We pre-computed it for 25 values of g and 13 rotation
 212 angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly
 213 (differentiably). The daily ratio s of the irradiance at the crop to that of the
 214 open field defines the crop’s shading. The canopy maximum temperature is
 215 lowered by $\kappa(1 - s)$.

216 κ is uncertain. Field studies under PV modules report small reductions
 217 of daytime crop or canopy temperature and little change in air temperature
 218 (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds
 219 calibrated on air temperature do not transfer directly to canopy temperature
 220 (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate

221 $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$
 222 for all methods (Section 3.2). Nocturnal warming under the modules is not
 223 represented. The reference evapotranspiration under the modules, ET_0 , is re-
 224 computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998)
 225 from the reduced radiation and canopy temperature. Soil water conserva-
 226 tion under the modules (Hassanpour Adeg et al., 2018; Elamri et al., 2018)
 227 therefore emerges from the model, but rain redistribution by the modules
 228 (Marrou et al., 2013a) is ignored.

229 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
 230 with the species parameters of its Table 1a (Table 1). Daily biomass growth
 231 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

232 where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as
 233 follows.

- 234 • f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends
 235 only on thermal time, with senescence accelerated by heat and drought
 236 stress. Shade therefore reduces the daily biomass increment but not
 237 leaf-area development.
- 238 • f_{Heat} decreases linearly between the heat threshold T_{heat} and the ex-
 239 treme temperature T_{ext} of the canopy maximum temperature.
- 240 • $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for
 241 drought (Woli et al., 2012) is computed from a daily soil-water balance
 242 with curve-number runoff, drainage and root water uptake. Irrigated
 243 fields have $f_{\text{Water}} = 1$.

244 SIMPLE has a fixed harvest index and therefore ignores the high sen-
 245 sitivity of grain number to shading around flowering (Fischer, 1985; An-
 246 drade et al., 1999). It also ignores the differences in shade tolerance be-
 247 tween crop types. In the meta-analysis of Laub et al. (2022), maize and
 248 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
 249 reals and tuber crops initially lost less than proportionally; shading trials
 250 also show marked effects of shade on maize development (Ramos-Fuentes
 251 et al., 2023). We added a stage-dependent penalty. During a critical window
 252 ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$
 253 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h

254 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
 255 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice
 256 0.5, for which no specific curve was available) and lowest for potato (0.2).
 257 These values are assumptions that reproduce this ranking; they are not cal-
 258 ibrated estimates. Section 3.6 shows the resulting shade-response curves.
 259 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
 260 ciency of shaded canopies under more diffuse light. The penalty, conversely,
 261 makes shading around flowering more damaging than proportional. The net
 262 effect is closer to the field evidence than either assumption alone.

263 The thermal-time requirement to maturity was calibrated for every site as
 264 the mean thermal time between the GGCM sowing and maturity dates under
 265 the 2001–2020 climate, and the canopy parameters were scaled accordingly.
 266 The cultivar is thus adapted to the present climate, and warming shortens the
 267 season. Growth stops smoothly once the thermal-time requirement is met.
 268 Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site
 269 is represented by a rainfed and an irrigated sub-field in proportion to its
 270 SPAM irrigated fraction. Each sub-field carries its own array, so both yield
 271 and electricity are area-weighted over the two sub-fields.

272 2.3. SALS: simulation-trained adaptive light sharing

273 SALS learns a *control network* $u_t = \pi_\omega(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 274 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 275 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 276 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 277 are the information available on day t :

- 278 • the crop and soil state: relative thermal time TT/T_{sum} , radiation in-
 279 terception f_{Solar} , relative soil water and the previous day’s ARID;
- 280 • a *forecast* of the day’s maximum temperature (relative to T_{heat} and
 281 T_{opt}), irradiance and ET_0 ;
- 282 • static information: the irrigation status of the sub-field, the array’s
 283 GCR g , latitude, day of season and the crop.

284 Forecasts are perturbed with random errors, independently every day: Gaus-
 285 sian with a standard deviation of 2 °C for maximum temperature, and mul-
 286 tiplicative with standard deviations of 20% for irradiance and 15% for ET_0 .
 287 The physics always uses the true weather. The same forecast errors are

288 applied to the rule-based controllers. The controller is conditioned on g .
 289 During training, each season is simulated with a GCR close to the density
 290 that is sustainable at its training site. This is the largest value on a coarse
 291 grid (0.05, 0.10, ..., 0.50) at which seasonal sharing keeps the mean retention
 292 of the site’s training seasons at or above ρ , plus a uniform perturbation in
 293 $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to con-
 294 trol the densities that design selection will later pick. Outside the cropping
 295 season the trackers backtrack for maximum electricity.

296 For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the
 297 yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional
 298 PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

299 which combines an exact (hinge) and a quadratic penalty for the food-security
 300 floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the
 301 exact gradient of Eq. (3) is obtained by back-propagation through the 365
 302 daily steps of every season (back-propagation through time). This is the key
 303 difference from reinforcement learning with black-box simulators (Gautron
 304 et al., 2022), which estimates gradients from sampled returns.

305 Each minibatch contains 512 site-seasons, each simulated as a rainfed and
 306 an irrigated sub-field, drawn from one of the four climates in turn. Seasons
 307 with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training
 308 and evaluation. The network is fine-tuned through the simulator with Adam
 309 (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm
 310 clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in
 311 a numerically stable form, so that gradients remain finite even under extreme
 312 heat.

313 *Initialisation.* Trained from random weights, the controller converged to
 314 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
 315 the seasonal-sharing rule, because the gradient of electricity with respect to u
 316 is small and of the same sign for most days of a season. We therefore initialise
 317 the network by behaviour cloning of the phenology rule of Section 2.4: binary
 318 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
 319 1.7 million day-states visited by the rule in the training seasons, with 1500
 320 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
 321 policy away from this initialisation wherever doing so improves the objective.
 322 Consequently, the gain of SALS over the phenology rule measures what fine-

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

323 tuning through the simulator adds to that rule, not what can be learned
324 without it.

325 *2.4. Deployable design selection and baselines*

326 A real array has one fixed density, which must be chosen before it is built.
327 Every method, SALS included, therefore uses the same *history-based design*
328 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
329 for which the method’s mean yield retention over the site’s 2001–2014 sea-
330 sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
331 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in
332 at least 80% of the historical seasons. The historical seasons of the test sites
333 are never used for training, and the weather of the evaluation years is never
334 used for design. We compare four controllers:

- 335 • *AV static*: backtracking trackers without light sharing ($u = 0$);
- 336 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
337 cropping season;
- 338 • *Stress rule*: an expert rule that shares light ($u = 1$) during the main
339 growth phase ($0.1 < \text{TT}/T_{\text{sum}} < 0.9$) except on days with a heat-stress

340 forecast ($T_{\max} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules
 341 track the sun and shade the crop;

- 342 • *SALS*: the learned controller.

343 The open field and a conventional PV plant without crops ($\text{LER} = 1$ each)
 344 are the references. Because the design rule is identical for all controllers, dif-
 345 ferences between methods reflect the quality of control and the array density
 346 that each controller can sustain under the food-security floor. We also com-
 347 pare the controllers at identical GCR over the whole grid (matched-density
 348 frontiers).

349 2.5. A shade-response model learned from field data

350 The simulator needs a crop response to shade that is supported by ob-
 351 servations. We learned one from the open data of the meta-analysis of Laub
 352 et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observa-
 353 tions of yield relative to an unshaded control at relative shade reductions of
 354 0–90% from 58 studies, with crop and crop type, experiment type (artificial
 355 shading or intercropping), climate zone, irradiation and shade type. Rela-
 356 tive yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$;
 357 maximum a posteriori estimation, so this is a hierarchical shrinkage regres-
 358 sion with bootstrap intervals and not a full Bayesian model). We compared
 359 six models: proportional loss $y = 1 - s$; one curve for all crops; one curve
 360 per crop type without pooling; one curve per crop type with partial pooling
 361 (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5);
 362 partial pooling with covariates on $\log \beta$ (standardised irradiation, subtrop-
 363 ical zone, solar-panel shade and intercropping); and gradient-boosted trees
 364 on the same inputs. Fits use L-BFGS. Models were compared by leave-one-
 365 study-out cross-validation, in which every study is held out once, which tests
 366 the prediction of a new study and not of a new observation in a known study.
 367 Intervals for the crop-type curves and covariate effects come from a cluster
 368 bootstrap over studies (150 resamples).

369 2.6. Shade-response uncertainty: gradient calibration, robust controllers and 370 design

371 The crop’s response to shade is the least certain part of the simulator,
 372 so we treat it as an unknown with a distribution. Two parameters per crop
 373 enter the simulator: a radiation-use compensation c (growth is multiplied by

374 $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop)
 375 and a multiplier h on the flowering-window harvest-index penalty. Because
 376 the simulator is differentiable, both can be fitted by gradient descent through
 377 whole simulated seasons to published yield-shading observations. The ob-
 378 servations are the relative yields at 20% and 40% shading of the data-driven
 379 curves of Section 2.5 for the five crops (median, with an error equal to the
 380 width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice
 381 takes the curve of C3 cereals, which contains no rice), the field trials on rice,
 382 wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-
 383 Fuentes et al., 2023) (each season of each trial as a separate observation, error
 384 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes
 385 et al., 2023). We call the resulting ensemble E2. A wider ensemble, E3, uses
 386 the same procedure with a target error of 0.2 for the learned curves, which
 387 is the between-study spread of a new study (Section 3.7); the spread of
 388 the resulting members is smaller (standard deviation of relative yield across
 389 members 0.04–0.09 at 20% shading and 0.05–0.12 at 40% shading, against
 390 0.02–0.04 and 0.03–0.06 for E2), because the fit also weights the field trials
 391 and the priors. An earlier ensemble, E1, was fitted in the same way to curves
 392 read by hand from the figures of Laub et al. (2022) and was used to train
 393 the robust and adaptive controllers below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and
 394 $h \sim \mathcal{N}(1, 0.5^2)$, and the parameters are clamped to $[0, 2.5]$ (c) and $[0, 2]$ (h)
 395 after every step, which is why a response milder than the data cannot be
 396 lowered further. We fitted 12 ensemble members by randomised maximum a
 397 posteriori estimation: every member sees the observations perturbed by their
 398 errors and a prior draw perturbed by its width, and is optimised with Adam
 399 (40 steps, with the loss still falling slightly at the last step; 40 training-block
 400 seasons per crop and member; shading of the static arrays cached). The
 401 ensemble is a device for generating plausible responses and not a validated
 402 posterior: it has no coverage check, and for some parameters (the harvest-
 403 index multiplier of wheat and potato) its spread equals that of the prior.
 404 The design analyses use five fixed members (indices 0, 3, 5, 8 and 10 of 12),
 405 chosen in advance. The ensemble is used in three ways. All training and
 406 evaluation ran on a 4-core CPU; one training run takes about 20–25 minutes
 407 and the calibration of one ensemble about 7 minutes.

408 *Robust and adaptive controllers.* SALS-robust has the architecture and
 409 training of SALS (Section 2.3), but every training season is simulated with the
 410 response of a randomly drawn ensemble member that the controller does not
 411 observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible

densities differ between members. SALS-adaptive additionally observes the realised biomass of the crop under the modules relative to an open reference plot, a quantity that can be measured on the farm by remote sensing or by a reference plot. A feedback rule that shares light whenever this ratio falls below a target serves as the hand-written counterpart; its target (1.0) was chosen on training-block seasons under the ensemble.

Uncertainty-aware design. The design rule of Section 2.4 chooses the largest GCR whose historical retention meets the floor. Under shade-response uncertainty we choose the largest GCR that meets the floor in at least 75% of five ensemble members, and compare it with a design under the ensemble mean and with an oracle design under the true response. Evaluation of this rule is leave-one-out: the truth is the held-out member and the design uses the other four.

Stress tests. Controllers and rules are also evaluated under four fixed responses, the proportional response, the response calibrated to the meta-analysis, the response fitted to the field trials and a harsh response (proportional growth and a 1.5 times larger harvest-index penalty). The proportional and harsh responses lie outside the range of the ensemble.

2.7. Economic analysis

To judge whether the electricity gains matter economically, we computed an annual net value per hectare for each method relative to the open field: electricity revenue minus the annuity and operating cost of the installed capacity minus the loss of crop revenue (crop production costs are unchanged). The installed capacity is 2.1 g MWp per hectare (modules cover a share g of the land at 0.21 kWp m⁻²). Yield is dry matter, valued at assumed prices. A conventional ground-mounted plant (GCR 0.4) that replaces the crop is the comparator. All parameters (Table 15) are assumptions chosen to be representative, not data, and their ranges drive the conclusions; we therefore report break-even electricity prices and one-at-a-time sensitivities, and we do not monetise food security, which market crop prices do not capture.

2.8. Evaluation protocol, statistics and upscaling

Sites were split spatially: 10°×10° blocks were randomly assigned to training (70%) or test (30%), so that no test site has a training site in the same block. Training uses the 2001–2014 seasons of the training sites. Design selection uses the 2001–2014 seasons of the test sites. Evaluation uses the

447 2015–2019 seasons of the test sites. The same site-seasons are analysed in all
448 four climates.

449 We report the following metrics:

- 450 • the mean electricity ($\text{MWh ha}^{-1} \text{yr}^{-1}$) and the relative electricity e ;
- 451 • the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- 452 • season-level compliance (share of seasons with $r \geq \rho$) and site-level
453 compliance (share of sites whose mean test-season retention is $\geq \rho$);
- 454 • the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- 455 • the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

456 Two aggregations are used and labelled in the text: tables report means over
457 all test seasons of a sample (season-pooled), whereas statistical comparisons
458 and intervals use the crop–site pair as the unit of analysis (mean of the pair’s
459 seasons, then mean over pairs), so the same contrast can differ slightly be-
460 tween the two (for example 0.034 and 0.038 for SALS minus the phenology
461 rule on the main split). Samples are also labelled: the main analysis uses
462 the 92 held-out pairs of one split and a GCR grid with step 0.025; analyses
463 pooled over three splits use the 76 distinct pairs of every third held-out pair
464 and a grid with step 0.05, and analyses of alternative shade responses on
465 one split use every sixth (21 pairs) or every fourth pair (22 pairs). Statis-
466 tical comparisons use site means as the unit of analysis. We report paired
467 Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence in-
468 tervals of the mean difference. SALS was trained with three random seeds
469 on the main split, and with one seed on each of two further spatial splits.

470 For the global estimate, every site is evaluated in the split in which it is
471 held out. The mean electricity and yield per hectare of each crop are scaled
472 to the *physical* area of that crop in SPAM 2010, which avoids double counting
473 multiple-cropped land (Waha et al., 2020). We report the installed capacity
474 (TWp), the capacity factor and a supply curve.

475 3. Results

476 3.1. Electricity and food on held-out sites and years

477 Table 2 compares the four controllers on the 92 held-out crop–site pairs of
478 the main split (95 pairs are held out; pairs without a season whose open-field

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in $^{\circ}\text{C}$, thermal times in $^{\circ}\text{C d}$, RUE in gMJ^{-1} ; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

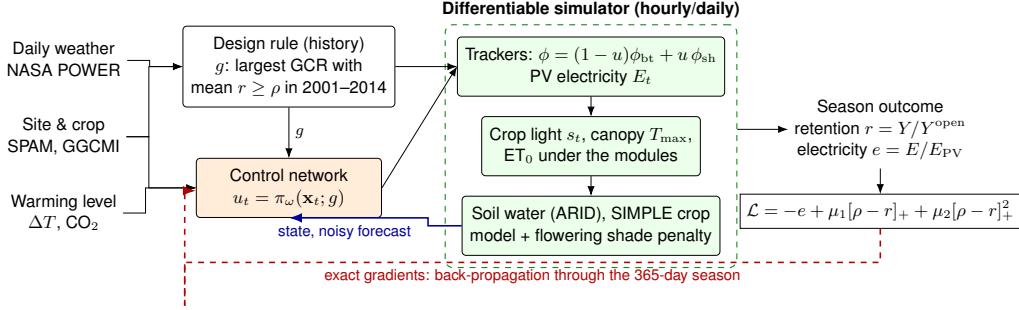


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

yield is at least 0.2 t ha^{-1} in all four climates are excluded) (seasons sown in 2015–2019), after each site’s density has been chosen from its 2001–2014 seasons with the same rule for all controllers. By construction all methods keep the mean yield retention close to 90% (0.904–0.917), so differences in electricity are differences in the density that each controller can sustain under this floor.

Static arrays are the weakest. Without light sharing, the floor limits the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$, 16% of a conventional PV plant per hectare (LER 1.07). Rotating the modules to let light through during the cropping season (seasonal sharing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule that shares light only while the crop is growing (phenology rule) reaches

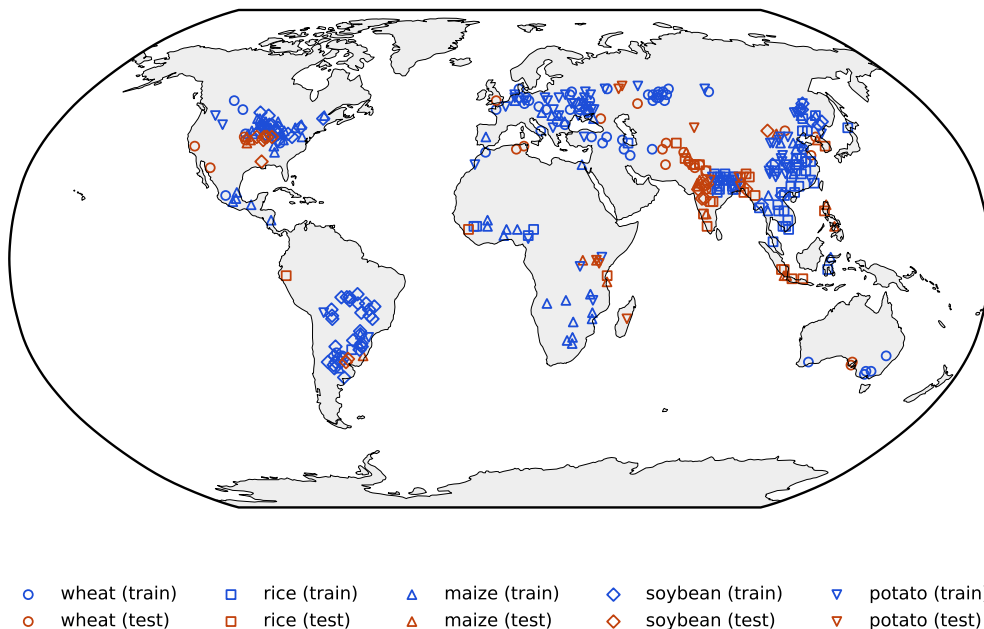


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

491 465 MWh ha⁻¹ yr⁻¹ (LER 1.24). An expert rule that additionally shades the
 492 crop on forecast heat or drought days (stress rule) is worse than the phenol-
 493 ogy rule (LER 1.18): its density is limited to 13% because it reduces light at
 494 times when the floor is most binding.

495 **The learned controller is best by a modest, consistent margin.**
 496 SALS sustains a density of 17% (similar to the phenology rule), produces
 497 510 MWh ha⁻¹ yr⁻¹ and reaches an LER of 1.270 at the same retention. Com-
 498 pared with the phenology rule this is +0.038 in LER (95% CI over sites
 499 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Ta-
 500 ble A.18). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35
 501 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows
 502 the same picture at matched densities: the learned policy lies on or slightly
 503 above the phenology rule over the whole density range, both lie well above
 504 seasonal sharing and the stress rule, and static arrays are far below.

505 **What the floor does and does not guarantee.** Enforcing the floor

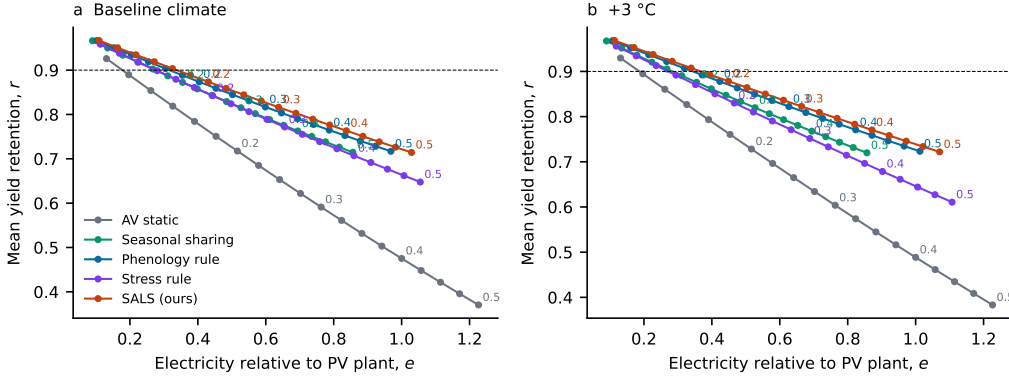


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

3.2. Robustness

Seeds and splits. Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at +3 °C. On two further spatial splits, in which different blocks of sites are held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at +3 °C (main split, season-pooled: 0.034 and 0.032; the site-level paired contrasts of Table A.18 are 0.038 and 0.034). In every split SALS ranks first, the phenology rule second and static arrays last.

Canopy cooling. The canopy-cooling coefficient κ is the most uncertain microclimate parameter. We repeated the whole protocol, including the historical design step, with $\kappa = 0, 1$ and 2 °C. Electricity changes with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and

528 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at
529 0.905 by the design rule and the ranking is unchanged. SALS exceeds the
530 phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate
531 (0.022, 0.032 and 0.051 at +3 °C).

532 **Which ingredients matter.** Two ablations test the design of SALS
533 at the same training budget. Masking the stress inputs (soil water, ARID,
534 and the maximum temperature, irradiance and ET₀ forecasts) leaves the
535 LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297
536 at +3 °C). Training only on the baseline climate gives 1.268 and 1.291. The
537 improvement over the rules therefore does not depend on stress-aware inputs
538 or on training in warmed climates.

539 *3.3. Floor conditioning, chance-constrained training and forecast errors*

540 **One network for all floors.** Regulations and farmers set different
541 floors. The network of the main experiments was trained at $\rho = 0.9$ only.
542 We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical
543 design step for each floor (Table 3). Its LER exceeds that of the phenology
544 rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037,
545 0.032 and 0.033 at +3 °C), so the benefit carries over to other floors without
546 retraining. A network that received the floor as an additional input and was
547 trained over all four floors did not improve on it (differences in LER ≤ 0.001).
548 Because the design step already adapts the density to the floor, conditioning
549 the controller on the floor is not needed.

550 **Layout, rotation limit and harvest-index assumptions.** We checked
551 three assumptions that could drive the ranking (Table 4; 21 pairs of the main
552 split, one training seed). The rotation limit of the trackers sets how much
553 light sharing can deliver: with a limit of 45° the LER of the phenology rule
554 is 1.138, with 60° 1.170 and with 80° 1.206, whereas static trackers stay at
555 1.05–1.06, so the benefit of light sharing grows with the limit and the ranking
556 is unchanged. Fixed non-tracking arrays held flat or tilted by 30° do not do
557 better than static trackers (LER 1.048 and 1.043 against 1.053), so the static
558 tracker is not a weak baseline relative to them; vertical bifacial layouts are
559 not modelled. The ranking of the rules also does not depend on the assumed
560 harvest-index penalty. Moving the critical window to 0.30–0.50 or 0.60–0.80
561 of the thermal-time requirement changes the LER of the phenology rule by
562 at most 0.001, and removing the penalty altogether raises all light-sharing
563 LERs (phenology rule 1.328, SALS 1.353) without changing the order: the
564 phenology rule exceeds seasonal sharing by 0.03 to 0.05 in every case (0.035

565 with the early window, 0.034 with the late window, 0.053 without penalty),
 566 because it produces electricity before canopy development and after maturity,
 567 when the crop does not use the light, and not because of the assumed
 568 window. The margin of SALS over the written rule changes with these assumptions
 569 (0.005 to 0.025 LER), and its margin over the tuned rule is 0.000
 570 to 0.009.

571 **A floor of 66%.** The German pre-standard DIN SPEC 91434 requires
 572 66% of the reference yield (DIN, 2021). Repeating the design step with
 573 $\rho = 0.66$ (baseline climate, main split, same network trained at 0.9) the
 574 ground coverage cap of 0.5 binds for the sharing methods (mean GCR 0.48
 575 for the phenology rule and 0.48 for SALS, 0.44 for the stress rule), static
 576 arrays reach a coverage of 0.24 and LER 1.28, the phenology rule 1.66 and
 577 SALS 1.71. At this floor the density is limited by the cap and by the mean
 578 retention of 0.73 reached at the cap, not by the floor, so the comparison of
 579 controllers says little about control; we do not interpret it further.

580 **Chance-constrained training.** We also trained a network whose ob-
 581 jective penalises a batch share of seasons below the floor that exceeds 20%.
 582 It did not improve the tail. With the mean-based design step its LER was
 583 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-
 584 constrained design it reached season-level compliance of 92.6% against 92.0%
 585 and an LER of 1.198 against 1.205. The season-level guarantee is obtained
 586 by the design step (Section 3.1), not by the training objective.

587 **Forecast errors.** We raised the forecast errors seen by the controllers
 588 to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and
 589 of the phenology rule was identical at all four error levels to three decimals
 590 (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and
 591 1.416). Neither controller uses the weather forecast: both rely on the crop’s
 592 thermal time, which is observable from on-farm temperature records. The
 593 stress rule, which depends on the forecast of maximum temperature, degrades
 594 with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and
 595 its LER from 1.432 to 1.418 as the errors triple.

596 **A tuned rule.** The phenology rule above was written, not tuned,
 597 whereas SALS is trained. We therefore tuned the window of the rule on
 598 the training-block seasons with the same objective (electricity at the density
 599 where mean retention equals the floor); the best window starts at 0.15 in-
 600 stead of 0.05 of the thermal-time requirement (Table 5). Pooled over three
 601 splits and 76 distinct crop–site pairs, the tuned rule raises LER over the orig-
 602 inal rule by 0.026 (95% cluster-bootstrap interval 0.016 to 0.037), and SALS

603 exceeds the tuned rule by 0.018 (0.012 to 0.024), or 5.7% more electricity at
 604 equal retention, and the original rule by 0.044 (0.030 to 0.060). Tuning the
 605 rule thus accounts for about 60% of the margin of SALS over the written
 606 rule. A five-parameter ramp-plateau schedule optimised by CMA-ES on the
 607 same training seasons performs like the tuned window (LER 1.238 against
 608 1.233, difference +0.005, interval -0.001 to $+0.010$), and SALS exceeds it
 609 by 0.013 (0.007 to 0.019); the optimum is an almost binary window (start
 610 0.17, end 0.93, level 0.99), so more flexible schedules of this family do not
 611 close the gap. The remaining margin of SALS is small but not zero; it is not
 612 explained by the tuning of a window or by a smooth ramp, and we did not
 613 test richer rule families (for example schedules that depend on soil water).
 614 SALS meets the floor in fewer seasons than the rules (83% against 84–85%).

615 *3.4. What the learned policy does*

616 Averaged over the rainfed test seasons of the main split at a density of 0.3,
 617 the light-sharing level of SALS is a smooth function of relative thermal time.
 618 It is close to zero in the first 10% of the thermal time required for maturity,
 619 rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and
 620 falls through 0.58 at 80–90% to about zero at maturity. The plateau covers
 621 the window in which our crop model penalises shading through the harvest
 622 index. The ramps at both ends let the modules produce electricity when
 623 additional light is worth least to the crop. Fig. 4 shows two example seasons.
 624 The phenology rule, which is also the initialisation of the network, shares
 625 light during the whole growing period. Fine-tuning through the simulator
 626 found that sharing can be reduced at the edges of the season: SALS gains its
 627 advantage by yielding light when it matters and collecting electricity when
 628 it matters less.

629 The policy also shares less light on days with a forecast heat stress: the
 630 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 ($+3^\circ\text{C}$)
 631 on other days, and 58–63% of sites show a lower level on heat days. This
 632 cannot be read as a stress response, however. Heat days occur late in the
 633 season, when the phenological ramp is already falling, and a policy without
 634 any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).
 635 The ablation above confirms that stress information does not improve the
 636 result.

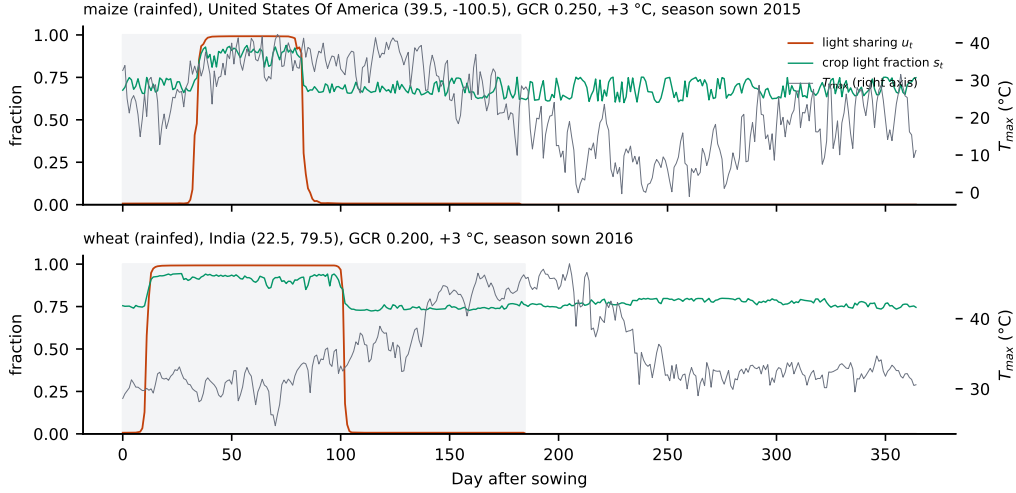


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

3.5. Crops and regions

The benefit of light sharing differs between crops (baseline climate, SALS). Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and 577 $\text{MWh ha}^{-1} \text{yr}^{-1}$; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31), and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20 for both). The stress rule has a lower LER for every crop (1.26 for soybean, 1.14–1.21 for the others).

The floor is not met evenly. Production-weighted retention is 0.888 for soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose more than low-yielding ones. Regionally, Latin America and Oceania (4 sites) combine the highest density (GCR 0.41) and electricity (1090 $\text{MWh ha}^{-1} \text{yr}^{-1}$) with a production-weighted retention of only 0.862. Africa and the Middle East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902 (67% of sites) at a density of 0.14. The regional figures rest on 4 to 39 pairs without intervals and should be read as indications. A regulatory floor enforced per site would therefore need a region- and crop-specific design step, not a global one.

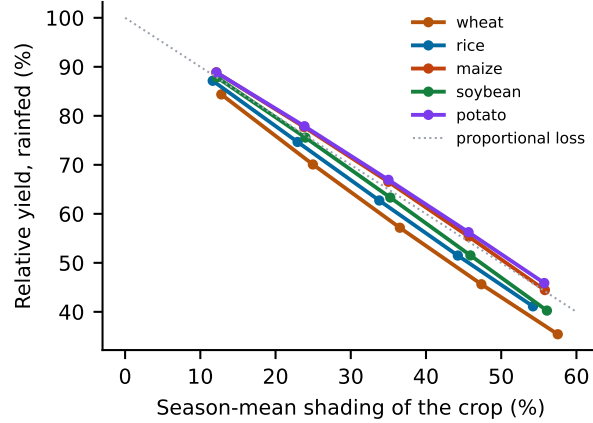


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

3.6. Model behaviour and validation

PV output. The simulated specific yield of an isolated tracker row agrees with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation of 5.5% (Section 2.2).

Shade response. Fig. 5 shows the simulated relative yield of rainfed crops under static arrays as a function of the season-mean shading of the crop. Yield declines close to proportionally with shading: 84–89% relative yield at 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat loses most and potato and maize least. This response is steeper than the *less than proportional* losses that Laub et al. (2022) report for small shade levels of C3 cereals and tubers, and it does not reproduce their finding that maize and grain legumes are more shade sensitive than C3 cereals. In the simulator, shading also relieves heat and drought in rainfed maize and soybean, which offsets the higher harvest-index penalty we assigned to them. The absolute effect of shading on each crop is therefore uncertain (Section 4.5).

Comparison with field trials. We fitted one radiation-use compensation coefficient per crop to the yield–shading curves of Laub et al. (2022) using training-block sites only (Methods, calibration script in the repository). The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the proportional model already lies within about 0.02 of the C3-cereal and tu-

ber curves, and it cannot lower the maize and legume responses below the proportional case (root-mean-square error 0.16 for maize and 0.08 for soybean). Against independent field trials (Table 6) the model underestimates retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al. (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table 7 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every method rise (phenology rule 1.28 against 1.23), and with a response fitted to the field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19 and +0.30 over static arrays (conservative, calibrated, field-fitted); only the order of the seasonal and stress rules changes. The absolute gains for rice, wheat and potato are thus conservative with respect to these trials; for maize and soybean the meta-analysis points the other way (Section 3.7).

3.7. *A shade-response model learned from field data*

Cross-validation by study shows what the data can and cannot support (Tables 8 and 9). Over all 339 shaded observations the root-mean-square error of a held-out study falls from 0.43 for proportional loss $y = 1 - s$ to 0.37 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes from crop types that the simulator does not represent (berries, forages, fruits and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses, the learned curves are not better than proportional loss: the error is 0.222 against 0.220 pooled (difference +0.002, 95% interval -0.023 to $+0.030$), 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes and tubers. Partial pooling is as good as separate curves (0.312 against 0.313) and the covariates add almost nothing (0.311). Gradient-boosted trees on the same inputs are worse (0.321) than the parametric model. The error of 0.22 for the four crop types is large: yields of a new study differ from the learned curve by about 20 percentage points, which is the between-study spread that a design under uncertainty has to cover. Crop type is the information that transfers between studies, and none of the covariate effects on $\log \beta$ is clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ; subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45).

713 The estimated effect of solar-panel shade is not interpretable, because all 46
 714 panel observations are of leafy vegetables, mainly lettuce, and the effect is
 715 confounded with crop type.

716 The learned curves show a clear ordering of crop types (Table 10). At
 717 20% shading C3 cereals and tubers lose less than proportionally (relative
 718 yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40%
 719 shading the values are 0.72 and 0.71 against 0.42 and 0.40. Intervals are wide
 720 for maize (0.27–0.59 at 40%) and for tubers, which rest on two studies. The
 721 simulator’s responses are close to the data for cereals and potato but, under
 722 both calibrated ensembles, too mild for maize and soybean at 40% shading
 723 (0.75 and 0.48 for E2 against 0.42 and 0.40 in the data). The calibration
 724 cannot lower the maize response further, because the simulated shading also
 725 relieves heat and drought stress and the compensation and harvest-index
 726 parameters are bounded below at zero. The field trials of maize are also
 727 milder than the meta-analysis: Ramos-Fuentes et al. (2023) report a harvest
 728 index unaffected by shade, and Sekiyama and Nagashima (2019) a yield 5.6%
 729 above the control under a low-density stilt-mounted array in one season at
 730 one site. For maize and soybean the direction of the bias is uncertain, and
 731 the gains reported in this paper are possibly overestimated.

732 3.8. *Learning under shade-response uncertainty*

733 **Calibration.** Gradient descent through the simulator fitted the five crops
 734 to the data-driven curves and the field trials (ensemble E2, Table 11). The
 735 data identify two features: the harvest index of maize is hardly affected by
 736 shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al.
 737 (2023), and soybean is more shade sensitive than the cereals (compensation
 738 0.05, harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the
 739 largest compensation (0.91 ± 0.31) and the largest uncertainty; the compen-
 740 sation is an effective parameter that absorbs diffuse-light and light-saturation
 741 effects that the simulator does not model explicitly, not a physical value.

742 **Learned controllers.** Controllers were evaluated under four fixed re-
 743 sponses on 21 crop–site pairs of split 0 with one training seed, so the intervals
 744 below, which are cluster bootstraps over pairs of per-pair mean LER differ-
 745 ences, do not include seed or split variation (Table 12 gives season means,
 746 which differ slightly). Under every response SALS, trained on the propor-
 747 tional response, is better than the written phenology rule: by +0.019 (0.007
 748 to 0.039), +0.024 (0.009 to 0.045), +0.038 (0.017 to 0.063) and +0.015 (0.000

749 to 0.036) LER under the proportional, calibrated, field-fitted and harsh re-
 750 sponses. Training on the narrow ensemble E1 did not change this: SALS-
 751 robust differs from SALS by -0.005 to $+0.003$ (intervals contain zero except
 752 for the proportional response, 0.000 to 0.003), and SALS-adaptive, which
 753 sees the measured biomass ratio, by -0.010 to $+0.007$ (all intervals contain
 754 zero). Training on the wider ensembles did not help either: SALS-robust on
 755 E2 and E3 and SALS-adaptive on E3 are below SALS by 0.014 , 0.017 and
 756 0.017 LER under the proportional response, 0.017 , 0.021 and 0.020 under the
 757 calibrated response, 0.004 , 0.005 and 0.011 under the field-fitted response (in-
 758 tervals contain zero) and 0.008 , 0.014 and 0.017 under the harsh response;
 759 the intervals exclude zero in 6 of the 9 comparisons for the proportional, cal-
 760 ibrated and harsh responses. The E3 robust controller is not distinguishable
 761 from the written rule ($+0.001$ to $+0.003$, intervals contain zero) except un-
 762 der the field-fitted response ($+0.032$, 0.001 to 0.067). A tuned window and a
 763 CMA-ES ramp schedule, also evaluated under the four responses (Table 12),
 764 are 0.005 to 0.018 LER below SALS (intervals of the SALS contrast: $+0.004$
 765 to $+0.016$ against the tuned window under the proportional response, and
 766 containing zero for the ramp under the calibrated and field-fitted responses
 767 and for the window under the harsh response), and the ensemble-trained
 768 controllers are at or below the tuned window under all responses except the
 769 field-fitted one. With one seed and 21 pairs these controllers therefore did not
 770 retain the margin of SALS within our training budget; a controller that must
 771 perform across plausible responses is less specialised, but a larger budget or
 772 more seeds could change this. The feedback rule is worse than the written
 773 rule by 0.026 – 0.058 LER under every response (intervals exclude zero). A
 774 plausible reason is that the biomass ratio does not reveal the harvest-index
 775 loss from shading around flowering, but we did not test this explanation.

776 **Design under uncertainty.** The larger risk is in the design step (Ta-
 777 bles 13 and 14, Fig. 6). The designs below are evaluated by leaving one
 778 ensemble member out as the truth and using the other four, so that the
 779 truth never enters the design. Pooled over the three spatial splits (76 dis-
 780 tinct crop-site pairs), a density chosen under the mean response of E2 leaves
 781 70% of the pairs at the floor when the true response is another plausible one,
 782 against 87% for an oracle design (E3: 71% and 85%). Choosing the density
 783 so that the floor is met in at least 75% of the other members raises compli-
 784 ance by $+7.9$ points under E2 (95% cluster-bootstrap interval 4.5 to 11.6)
 785 and $+7.9$ under E3 (3.9 to 12.4), at a cost of 0.031 and 0.025 in LER. Raising
 786 the floor used in the design from 0.90 to 0.92 under the mean response does

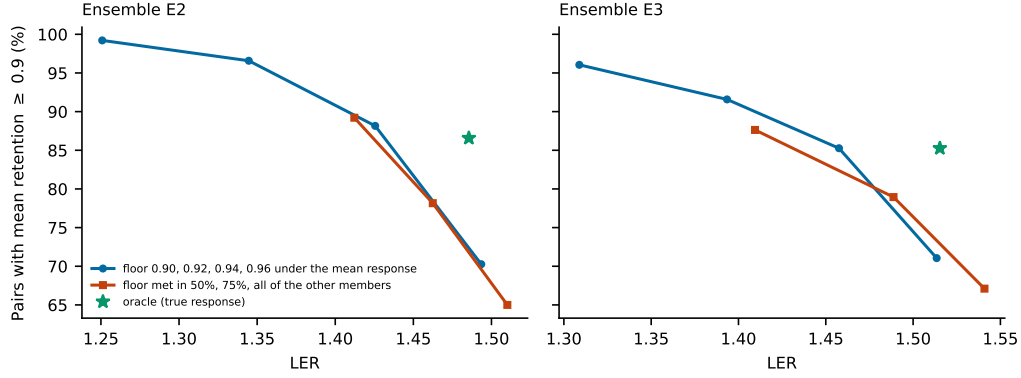


Figure 6: Share of crop-site pairs at the floor against LER for the phenology rule under design rules for shade-response uncertainty (leave-one-member-out, three splits, 76 pairs): a higher floor under the mean response (blue; the first point is the design under the mean response) and a floor met in a share of the other members (orange), with the oracle design (green).

more: +18.0 points (13.4 to 22.6) at 0.068 LER under E2 and +14.2 points (9.7 to 19.2) at 0.056 under E3, and the pairs below the floor fall short by less (mean shortfall 0.0022 against 0.0045 for the quantile rule under E2). The two kinds of rule lie on one frontier (Fig. 6): at about 88–89% compliance the ensemble rule that requires all other members to meet the floor gives LER 1.412 under E2 against 1.426 for a floor of 0.92, and under E3 1.409 against 1.458, so a simple margin is as good as or better than the ensemble rule by 0.01 to 0.03 LER. The ensemble is thus useful for sizing the margin (0.02 to 0.04 above the floor for 85–97% of the pairs), not for replacing it. None of these rules is safe outside the ensemble (Table 14): under the proportional, calibrated and harsh responses the mean-response design leaves 7–17% of the pairs at the floor, a floor of 0.92 or the ensemble rules 12–33%, a floor of 0.94 29–47% and a floor of 0.96 51–78%, at a cost of about 0.2 LER, whereas the field-fitted response, which lies inside the range spanned by the ensembles, is handled well by every rule (80–99%) except the mean-response design (68–79%). The guarantee therefore holds only within the range of responses that the ensemble spans, and a response harsher than the data supported by the ensemble breaks it.

3.9. Economics of light sharing

All figures in this section follow from the assumed parameters of Table 15 and should be read as a sensitivity analysis, not a forecast. The economics are dominated by the capacity installed, not by the crop (Table 16): the annuity of the modules is 11–32 kUSD per hectare and year, whereas the crop revenue lost is about 0.3 kUSD, so a method is worth what it earns per installed watt. With the densities designed for the floor, static arrays earn +0.3 kUSD ha⁻¹ yr⁻¹ at 60 USD MWh⁻¹ and break even at 59, whereas all light-sharing methods lose money at that price (–3 to –10 kUSD) and break even at 70 (stress rule), 73 (SALS), 77 (tuned phenology rule), 80 (written rule) and 92 (seasonal sharing) USD MWh⁻¹. At 90 USD MWh⁻¹ static arrays (6.3 kUSD) are better than the written (4.0) and tuned (5.5) rules and worse than SALS (7.6), which overtakes static arrays above about 85 USD MWh⁻¹. A conventional plant that replaces the crop breaks even at 48 USD MWh⁻¹ and is more profitable than every agrivoltaic option above that price (16 kUSD at 60, 94 kUSD at 120, against –6 and 21 for SALS); below it all options lose money and static arrays lose least (at 30 USD MWh⁻¹: plant –23.2, SALS –18.7, static –5.7 kUSD). The break-even price of SALS exceeds that of the plant by about 25 USD MWh⁻¹, which is the order of the premium or the value of food production that agrivoltaics needs for parity with the open field; parity with the plant needs more. The sensitivity is large (Table 17): at 90 USD MWh⁻¹ the net value of SALS changes from +17 to –8 kUSD between module costs of 0.7 and 1.5 USD Wp⁻¹, whereas crop prices matter little.

Because the density was designed for the floor and not for profit, we also let each site choose, among the densities that meet the floor in its history, the one that maximises its historical net value (sites where none pays stay open field; Fig. 7). Then no method pays at 30–40 USD MWh⁻¹; at 60 14–59% of the pairs install an array (none for seasonal sharing) and the net values are small (static arrays 0.9, SALS 0.3 kUSD); static arrays are the best option up to about 75 USD MWh⁻¹, SALS leads from about 80 (4.9 against 4.3 at 80, 8.3 against 6.3 at 90, 20.7 against 12.3 at 120), and the tuned rule overtakes static arrays only above about 105. The value of control is therefore real but depends on a high price: at 90 USD MWh⁻¹ SALS is worth 1.7 kUSD ha⁻¹ yr⁻¹ over the tuned rule with profit-optimal densities.

Cost of the safety margin. At 90 USD MWh⁻¹, moving from a density designed under the mean response to a floor of 0.92 under that response costs 1.00 kUSD ha⁻¹ yr⁻¹ (17%) under E2 for 18 more points of compliance (70%

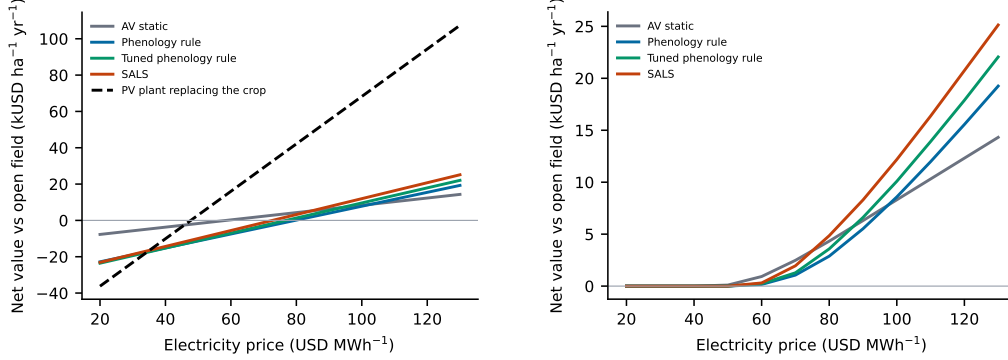


Figure 7: Net value per hectare relative to the open field against electricity price. Left: densities designed for the floor, with a conventional plant replacing the crop (dashed). Right: each site chooses the feasible density that maximises its historical net value. Illustrative parameters of Table 15.

843 to 88%) and 1.13 kUSD (20%) under E3 for 14 points (71% to 85%), about
844 56 and 79 USD ha⁻¹ yr⁻¹ per point. The ensemble rule that requires all other
845 members to meet the floor costs 0.85 and 1.22 kUSD for 19 and 17 points
846 (45 and 74 USD per point), so on this measure it is cheaper than the margin
847 under E2 and similar under E3. At 60 USD MWh⁻¹ a margin has a negative
848 cost, because every module beyond break-even loses money; the economics
849 of the design margin therefore depend on the price level, whereas its effect
850 on compliance does not.

851 3.10. Agrivoltaics under warming

852 Under warming, the open-field production of the held-out sites (Fig. 8)
853 falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soy-
854 bean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production
855 with light-sharing control follows the open field at about 90% of its level, so
856 it inherits the same climate impacts (maize -31%, rice -27% relative to the
857 baseline open field at +3 °C). Agrivoltaics does not protect staple production
858 against warming in this model. The LER of light-sharing systems increases
859 slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265),
860 because the floor allows slightly denser arrays in warmer conditions (GCR
861 0.173 to 0.177 and electricity 510 to 540 MWh ha⁻¹ yr⁻¹). The margin of
862 SALS over the rules does not grow with warming (Table A.18).

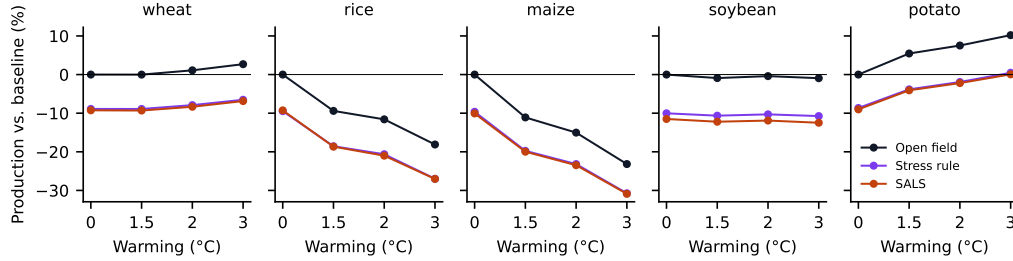


Figure 8: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

3.11. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop–site pairs, in at least one split; 222 are held out in at least one split and three have no valid season), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr^{-1} under the baseline climate and 16.0 PWh yr^{-1} at $+3^\circ\text{C}$, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr^{-1} , and static arrays 6.4 and 6.9 PWh yr^{-1} . For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWP of modules (11.3 TWP at $+3^\circ\text{C}$), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 9): the best 2% of the staple cropland would supply 16 PWh yr^{-1} and the best 5% 40 PWh yr^{-1} with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also among the regions where the floor is hardest to meet (Section 3.5). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

Maize and soybean, for which the simulated shade response is probably too mild (Section 3.7), account for 53% of the electricity of the upscaling

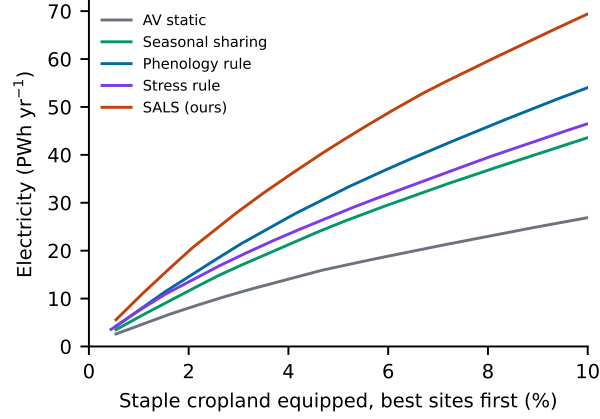


Figure 9: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

889 (maize 30%, soybean 23%, wheat 29%, rice 15%, potato 3%). Counting only
890 wheat, rice and potato would give about 7 PWh yr⁻¹ at the same coverage,
891 so the global figure should be read as an upper estimate for about half of its
892 content. By crop, the LER of SALS under the baseline climate is 1.20 for
893 wheat and rice, 1.31 for potato, 1.30 for maize and 1.41 for soybean.

894 4. Discussion

895 4.1. What the results establish

896 Four findings stand on consistent evidence. First, sharing light with the
897 crop is the main lever on the electricity that tracker-based agrivoltaic arrays
898 can produce under a yield floor. With static trackers the floor limits arrays
899 to about 6% ground coverage and an LER of 1.07. Any light-sharing control
900 allows 2.1 to 2.9 times the density, and the LER rises to 1.18–1.24 for hand-
901 written rules. Second, the best simple rule follows crop phenology. It shares
902 light across the middle of the season, around flowering and grain filling, and
903 produces electricity at the beginning and the end. It needs only the crop's
904 thermal time, which is observable on the farm, and it is insensitive to forecast
905 errors, whereas the rule that reacts to heat and drought stress is worse in
906 every climate and degrades with forecast error. Third, the qualitative result
907 does not depend on the weakest part of the model. The simulated shade

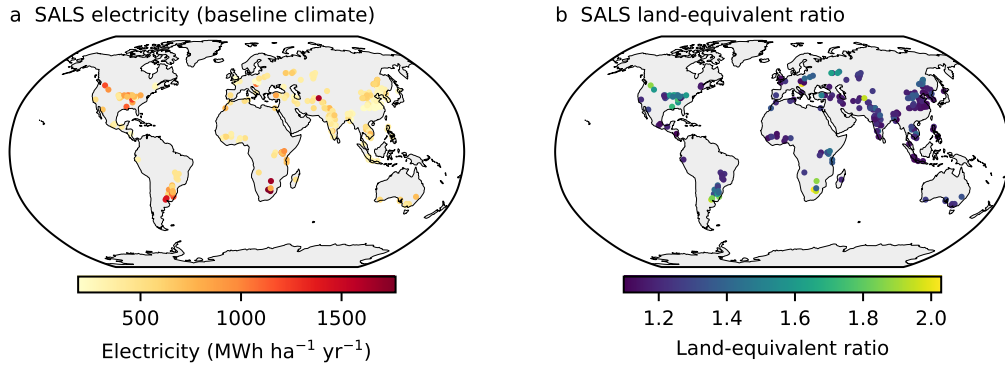


Figure 10: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

908 response is conservative against the field trials we could use (Section 3.6);
 909 calibrating it raises the allowed density, and the phenology rule stays the
 910 best method, with a gain over static arrays of 0.17, 0.19 and 0.30 LER
 911 under the conservative, calibrated and field-fitted responses. However, the
 912 response learned from the meta-analysis shows steeper losses for maize and
 913 grain legumes than the simulator reproduces (Section 3.7), so the absolute
 914 gains are conservative for wheat, rice and potato and probably overestimates
 915 for maize and soybean (53% of the upscaled electricity). Fourth, a controller
 916 learned through the differentiable simulator is better than the best rule we
 917 could write, by 0.04 in LER against the written rule (0.018 against a rule
 918 whose window was tuned on the training seasons, 5.7% more electricity at
 919 equal retention), reproducibly over training seeds (range 0.002 in LER), three
 920 spatial splits, three canopy-cooling assumptions and four climates, and at 84–
 921 86% of the held-out sites. This gain is small compared with the gap between
 922 static and dynamic control.

923 4.2. What role does artificial intelligence play?

924 We tested three ways in which learning could matter, and the answers dif-
 925 fer. The first is learning the crop’s response to shade from data. The learned
 926 model is useful mainly as a measurement of how little can be learned: a hier-
 927 archical model fitted to 428 observations from 58 studies predicts a held-out
 928 study of the four staple crop types with an error of 0.22 in relative yield, no
 929 better than proportional loss, and the gain over proportional loss (0.43 to

0.31 over all types) comes from crop types that the simulator does not represent. Crop type is the information that transfers, whereas irradiation, climate zone and shade type add nothing measurable and a non-parametric model (gradient-boosted trees) is worse with this number of studies (Section 3.7). Two things follow. The shade response of the staple crops is a distribution with a spread of about 20 percentage points between studies and not a curve, and the process-based simulator does not reproduce the steep losses of maize and legumes seen in the data. Learning the response was limited by the data and not by the model: two studies support the tuber curve, no rice study exists, and 46 panel observations are all leafy vegetables.

The second is calibration through the differentiable simulator. Gradient descent through whole seasons fitted the response of each crop to the learned curves and to the field trials, and it produced an ensemble of plausible responses that reconciles them (Section 3.8). The ensemble informs the design decision: a density chosen under the mean response leaves about 70% of crop–site pairs at the floor if another plausible response holds, against 85–87% for an oracle, and a density chosen to meet the floor in 75% of plausible responses recovers about half of that gap at a cost of 0.04 LER. A simple margin, however, does as well: raising the floor from 0.90 to 0.92 under the mean response reaches 85–88% of the pairs, and at equal compliance it keeps 0.01 (E2) to 0.03 (E3) more LER than the ensemble rule that requires all other members to meet the floor, while the cost per point of compliance is similar. None of the rules is safe outside the ensemble (7–33% of the pairs at the floor under the proportional, calibrated and harsh responses for the mean design, the margin of 0.92 and the ensemble rules), and only a floor of 0.96 reaches 51–78% at a cost of about 0.2 LER. What the ensemble adds is the size of the margin, not a better rule. The quantile rule is a scenario-type design (Calafiore and Campi, 2006; Campi and Garatti, 2008); with five members it has none of the finite-sample guarantees of that theory, and the comparison with the margin shows that it is not needed in practice. The bottleneck of tracker-based agrivoltaics under a food-security floor is the uncertainty of the crop response that enters the design, and it is data on that response, not larger networks, that would improve the result; the learned curves show a spread of about 20 percentage points between studies. The guarantee holds only within the range of responses that the ensemble spans.

The third is control, and here learning contributes little. The learned controller adds 0.018 LER over a tuned phenology rule (and 0.044 over the written one), and 0.015–0.038 over the written rule under the four alternative

968 responses on one split (21 pairs); a reader who needs a simple, auditable
 969 controller can use that rule and obtain most of the electricity gain over static
 970 arrays. Neither of the two extensions that we expected to matter under
 971 uncertainty did: training on ensembles of responses (E1, E2 and E3) did not
 972 help and, for the wider ensembles, did not retain the margin over the written
 973 rule within our training budget (one seed, 21 pairs), and neither did giving
 974 the controller a farm-measurable signal, the biomass of the crop relative to
 975 a reference plot, nor a feedback rule built on that signal. The learned policy
 976 also has to be initialised from the phenology rule (Section 4.5). We therefore
 977 do not claim that learned control is what makes agrivoltaics work: in this
 978 model, the schedule that a learned policy converges to is a smooth function
 979 of crop phenology, which a rule can already express.

980 *4.3. What the results do not establish*

981 The early hypothesis that stress-aware control, that is, shading the crop
 982 on heat and drought days, would be a main source of benefit under warming
 983 was not supported. The stress rule performs worse than the phenology rule
 984 in every climate, and a learned controller without stress inputs performs as
 985 well as one with them. A plausible reason is that in our crop model the yield
 986 gain from shading on a hot or dry day is smaller than the loss of light on
 987 that day, so stress information does not help. Whether this holds for crop
 988 models with more detailed canopy-temperature and heat-stress responses is
 989 an open question. Agrivoltaics does not protect staple production against
 990 warming in our simulations: light-sharing systems produce about 90% of the
 991 open-field yield in every climate and follow the same decline in maize and
 992 rice (Fig. 8).

993 *4.4. Food security in practice*

994 Holding the historical mean retention at 90% does not hold it in each
 995 season or at each site. 70% of SALS seasons meet the floor, the lower tail is
 996 slightly heavier than for the rules, and production-weighted retention is below
 997 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-
 998 constrained design raises season-level compliance to about 92% at a cost of
 999 0.065 in LER, and the ranking of the controllers is preserved. Whichever for-
 1000 mulation is chosen, a per-site regulatory floor would need a region- and crop-
 1001 specific design step. The upscaled potential (Section 3.11) shows the trade-off
 1002 in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply
 1003 need about 10 TWp of modules, several times the present global fleet, and

the best sites are rainfed maize and soybean areas where production-weighted retention is lowest.

4.5. Limitations

- *Simulation only.* All results come from a simulator. PV output was benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop response to shade was learned from one meta-analysis data set and compared with four field trials (Sections 3.6 and 3.7). The simulated response is conservative with respect to the trials (rice, wheat, potato, maize) but too mild for maize and soybean relative to the meta-analysis, where the 40%-shading relative yields are 0.42 and 0.40 against 0.75 and 0.48 in the simulator. The harvest-index sensitivities h are assumptions, and the sensitivity analyses use subsamples of crop-site pairs (22 for the rule comparison under alternative responses, 21 for the controller comparison, one training seed).
- *Canopy cooling.* The effect of shade on canopy temperature was represented by one coefficient κ , applied to a heat-stress function that was calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018). The ranking of the controllers is robust to κ between 0 and 2°C , but the absolute electricity changes by about $\pm 8\%$. Nocturnal warming under modules and rain redistribution are not modelled.
- *Crop model.* SIMPLE has a fixed harvest index and linear radiation-use efficiency, and its CO_2 response is that of controlled experiments that tend to overstate field responses (Long et al., 2006). More detailed models with grain-number formation and canopy energy balance may favour different schedules.
- *Layouts and models.* The baseline of static arrays is a backtracking tracker held at the energy-optimal angle; flat and 30° -tilted fixed arrays did no better in a check on 21 pairs, but vertical bifacial layouts (Willockx et al., 2023; Campana et al., 2021) are not modelled, and the ground coverage ratio is a coarse proxy of crop productivity (Dupraz, 2023). The PV model is isotropic in the diffuse component and has no bifacial gain; it was benchmarked against PVGIS at low density only and has not been compared with a reference implementation such as pvlib (F. Holmgren et al., 2018). The rotation limit was varied between

1038 45 and 80 degrees and changes the size of the benefit but not the rank-
 1039 ing; the harvest-index window is assumed to be common to all crops,
 1040 and shifting it or removing the penalty did not change the ranking of
 1041 the rules.

- 1042 • *Climate scenarios.* Warming was applied as uniform temperature shifts
 1043 with fixed precipitation, radiation and relative humidity, and with un-
 1044 changed cultivars and sowing dates. It is not a bias-adjusted climate
 1045 projection.

- 1046 • *Initialisation and state information.* The network is initialised from the
 1047 phenology rule, because training from random weights collapsed to full
 1048 sharing. The gain of SALS is therefore the gain of fine-tuning that rule,
 1049 and we cannot say whether a better optimiser or a richer policy class
 1050 would find larger gains. The controller uses the simulated state of the
 1051 crop and soil (thermal time, canopy interception, soil water) without
 1052 measurement error, and the forecast errors we tested apply to weather
 1053 only.

- 1054 • *Economics and deployment.* The economic analysis (Section 3.9) uses
 1055 assumed prices and costs, no grid, curtailment, storage or subsidy
 1056 model, and does not value food security; its break-even prices move
 1057 with the capex and the discount rate (Fig. 7). The value of midday
 1058 electricity, curtailment, storage, structural cost and farmer acceptance
 1059 (Trommsdorff et al., 2021) will determine the realisable share. The av-
 1060 erage row pitch at the chosen densities (about 13 m for 2.2-m modules)
 1061 leaves room for machinery, but the densest designs (GCR up to 0.5)
 1062 have a pitch of only 4.4 m.

1063 5. Conclusions

1064 Light sharing, the daily choice between maximum electricity and letting
 1065 light through to the crop, is the main lever of tracker-based agrivoltaics under
 1066 a food-security floor. In a differentiable simulator of trackers, PV output,
 1067 crop light, microclimate and crop growth, evaluated on held-out crop-site
 1068 pairs (92 on the main split, 76 distinct pairs pooled over three splits for the
 1069 controller and design comparisons) with the array density of each site chosen
 1070 from its own history, light-sharing rules allowed 2.1 to 2.9 times the module
 1071 density of static arrays at a mean yield retention of 90%. They raised the

1072 land-equivalent ratio from 1.07 to 1.18–1.24, and a rule that follows crop
 1073 phenology and needs no weather forecast was the best of them. The simu-
 1074 lated shade response is conservative against the field trials we could use for
 1075 rice, wheat and potato, and calibrating it raised the gains without chang-
 1076 ing the ranking of methods; for maize and soybean, which supply 53% of
 1077 the upscaled electricity, the meta-analysis indicates larger losses than the
 1078 simulator reproduces whereas one maize trial indicates smaller ones, so the
 1079 gains for these crops are uncertain and possibly overestimated. The crop’s
 1080 response to shade, the least certain input, was learned from 428 field obser-
 1081 vations: a hierarchical model predicted a held-out study of the staple crop
 1082 types no better than proportional loss (error 0.22), which shows the size of
 1083 the between-study spread, and gradient descent through the simulator turned
 1084 it into an ensemble of plausible responses. A learned controller exceeded the
 1085 written phenology rule by 0.044 LER and a rule with a window tuned on the
 1086 training seasons by 0.018 (95% interval 0.012–0.024), by following a smooth
 1087 phenology-dependent schedule that a rule can express; training it on ensem-
 1088 bles of shade responses or giving it a farm-measurable biomass signal did
 1089 not help, and with the wider ensembles it did not retain its margin over the
 1090 written rule in our training budget (one seed, 21 pairs). The decision that
 1091 mattered was the design. A density chosen under the mean response left
 1092 about 70% of crop–site pairs at the floor when another plausible response
 1093 held (oracle 85–87%); choosing the density so that the floor holds in 75%
 1094 of the plausible responses recovered about half of the gap at a cost of 0.04
 1095 LER, but a floor raised by 0.02 under the mean response was as good as or
 1096 slightly better at equal compliance, and no rule was safe outside the ensem-
 1097 ble. The ensemble is therefore useful to size a safety margin, not to replace
 1098 it. Under illustrative costs, light sharing breaks even with the open field at
 1099 73 (learned controller) to 92 USD MWh^{−1}; a conventional plant replacing the
 1100 crop stays more profitable at every price, so agrivoltaics depends on a pre-
 1101 mium, land-use rules or a value for food production that markets do not give.
 1102 The yield floor is met on average but not in every season or region, and the
 1103 electricity potential of 5% of staple cropland (about 14 PWh yr^{−1}) requires
 1104 about 10 TWp of modules. Future work should validate the crop response
 1105 to shade under trackers with more field data, notably for maize and legumes
 1106 where the meta-analysis and the trials disagree, extend the crop model to
 1107 grain-number formation and canopy energy balance, add costs and grid con-
 1108 straints, and pursue controllers that guarantee the food-security floor per site
 1109 and per season.

1110 **CRedit authorship contribution statement**

1111 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal
1112 analysis, Investigation, Data curation, Visualization, Writing – original draft,
1113 Writing – review & editing.

1114 **Declaration of competing interest**

1115 The author has no known competing financial interests or personal re-
1116 lationships that could have appeared to influence the work reported in this
1117 paper.

1118 **Data and code availability**

1119 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
1120 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA
1121 POWER daily weather (Sparks, 2018) the SIMPLE crop parameters (Zhao
1122 et al., 2019) and the shade-response meta-analysis data of Laub et al. (2022)
1123 (Zenodo 10.5281/zenodo.5716091, CC BY 4.0, copied into the repository).
1124 The code of the simulator, of SALS and of all experiments, together with the
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1126

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1130 **Appendix A. Statistical comparison**

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Sensitivity of the controller comparison to the rotation limit of the trackers and to the harvest-index penalty (LER; main split, every sixth held-out pair, 21 pairs, main shade response, history-based design on a GCR grid with step 0.05). SALS was trained for the main setting. Fixed arrays (no tracking; flat and tilted by 30° from the horizontal about the north–south axis) reach LER 1.048 and 1.043 in the same sample, against 1.053 for static trackers.

Setting	Static tracker	Phenology rule	Tuned rule	SALS
Main (limit 60°, window 0.45–0.65)	1.053	1.170	1.180	1.189
Rotation limit 45°	1.057	1.138	1.144	1.144
Rotation limit 80°	1.053	1.206	1.219	1.224
Window 0.30–0.50	1.053	1.169	1.176	1.185
Window 0.60–0.80	1.053	1.171	1.181	1.182
No harvest-index penalty	1.109	1.328	1.347	1.353

Table 5: Rules and the learned controller pooled over the three spatial splits (76 distinct crop-site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05, coarser than the 0.025 grid of the main analysis). The tuned phenology rule shares light while thermal time lies between 0.15 and 1.0 of the season requirement (window chosen on training-block seasons of split 0; the written rule uses 0.05 to 1.0). The optimised ramp schedule has five parameters (start 0.17, end 0.93, ramp widths 0.02, level 0.99) found by CMA-ES on the same training seasons.

Method	GCR	E/E_{PV}	Retention	Seasons ≥ 0.9 (%)	Pairs ≥ 0.9 (%)	LER
AV static	0.059	0.153	0.922	95	96	1.076
Seasonal sharing	0.151	0.249	0.915	84	91	1.164
Stress rule	0.118	0.255	0.918	83	88	1.173
Phenology rule	0.154	0.292	0.916	85	91	1.207
Tuned phenology rule	0.161	0.317	0.916	84	89	1.233
Optimised ramp schedule	0.155	0.322	0.916	83	87	1.238
SALS	0.161	0.335	0.916	83	87	1.251

Table 6: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 7: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 8: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table 9: Leave-one-study-out prediction error (RMSE of relative yield, observations with $s > 0$) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

Group	Obs./studies	Proportional	Hierarchical	Boosted t
C3 cereals	30/10	0.214	0.187	0.205
Maize	40/10	0.225	0.235	0.248
Grain legumes	58/14	0.215	0.225	0.234
Tubers/root crops	10/2	0.250	0.254	0.257
Four crop types used by the simulator	138/30	0.220	0.222	0.234
Other types (berries, forages, fruits, vegetables)	201/29	0.528	0.361	0.369
All types	339/58	0.430	0.312	0.321

Table 10: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 11: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table 12: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out pair, 21 pairs, coarse GCR grid, one training seed; season means, so differences differ slightly from the pair-level intervals in the text). SALS is trained on the proportional response; robust controllers are trained on random members of ensemble E1, E2 or E3 (adaptive: with the measured biomass ratio). The proportional and harsh responses lie outside the ensembles.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule	1.164	1.190	1.456	1.122
Tuned phenology rule	1.170	1.199	1.472	1.129
Optimised ramp schedule	1.170	1.199	1.477	1.122
SALS	1.178	1.209	1.489	1.132
SALS-robust, E1	1.179	1.204	1.491	1.127
SALS-adaptive, E1	1.176	1.202	1.495	1.124
SALS-robust, E2	1.165	1.192	1.483	1.126
SALS-robust, E3	1.163	1.191	1.481	1.121
SALS-adaptive, E3	1.164	1.191	1.475	1.118

Table 13: Array design under shade-response uncertainty, phenology rule, three spatial splits pooled (76 distinct crop–site pairs, every third held-out pair). Truth: each of five members of the ensemble in turn; the mean response and the member-based rules use the other four members (leave-one-out, so the truth never enters the design). E2: ensemble calibrated to the learned curves and field trials; E3: same procedure with target errors of 0.2 on the learned curves. Pairs: share with mean retention ≥ 0.9 ; shortfall: mean of $\max(0, 0.9 - \text{retention})$ per pair. The LER values exceed those of the main analysis because E2 and E3 are milder than the proportional response and allow denser arrays; the margin grid (0.92, 0.94, 0.96) was chosen after a first analysis of E2 and is not pre-specified.

Design	E2			E3		
	Pairs ≥ 0.9 (%)	Shortfall	LER	Pairs ≥ 0.9 (%)	Shortfall	LER
oracle (true response)	87	0.0025	1.486	85	0.0029	1.486
mean response	70	0.0062	1.493	71	0.0093	1.493
floor met in 50% of other members	65	0.0084	1.510	67	0.0125	1.510
floor met in 75% of other members	78	0.0045	1.462	79	0.0083	1.462
floor met in all other members	89	0.0019	1.412	88	0.0032	1.412
floor 0.92 under mean response	88	0.0022	1.426	85	0.0051	1.426
floor 0.94 under mean response	97	0.0007	1.345	92	0.0027	1.345
floor 0.96 under mean response	99	0.0003	1.251	96	0.0015	1.251

Table 14: Design rules outside the ensemble: share of crop–site pairs (%) with mean retention ≥ 0.9 under four fixed responses (P proportional, C calibrated to the meta-analysis, F field-fitted, H harsh), phenology rule, three splits pooled (76 distinct pairs). Designs use the whole ensemble E2 or E3 (five members and their mean); the oracle uses the true response. LER in the last column is the mean over the four responses.

Design	E2				E3				LER	
	P	C	F	H	P	C	F	H	E2	E3
oracle (true response)	91	89	87	84	91	89	87	84	1.283	1.283
mean response	13	13	68	12	13	17	79	7	1.435	1.448
floor met in 75% of members	16	16	80	12	22	25	93	13	1.384	1.397
floor met in all members	17	17	80	12	29	32	93	16	1.348	1.336
floor 0.92 under mean response	17	20	86	12	29	33	91	20	1.376	1.399
floor 0.94 under mean response	45	47	97	34	45	45	97	29	1.303	1.345
floor 0.96 under mean response	63	78	99	54	58	63	99	51	1.220	1.264

Table 15: Assumed economic parameters (illustrative, not data). Ranges used in the sensitivity analysis are in parentheses.

Parameter	Value
Electricity price	60 USD MWh ⁻¹ (30–120)
Capex, AV single-axis tracker plant	1.0 USD Wp ⁻¹ (0.7–1.5)
Capex, conventional tracker plant	0.75 USD Wp ⁻¹
Discount rate, lifetime, O&M	6% (4–8%), 25 years, 1.5% of capex per year
Installed capacity	2.1 MWp per hectare of module area share g (0.21 kWp m ⁻²)
Crop price per tonne of dry matter	wheat 290, rice 400, maize 250, soybean 520, potato 1000 USD (0.

Table 16: Illustrative annual economics per hectare (76 crop–site pairs, three splits, main shade response, baseline climate; parameters in Table 15). Net value is relative to the open field (crop costs unchanged) in kUSD ha⁻¹ yr⁻¹ at three electricity prices; break-even is the price at which the net value is zero. The conventional plant replaces the crop.

Method	E (MWh)	Capex annuity	Crop loss	Net at 60	Net at 90	Net at 120
AV static	200	11.5	0.27	0.3	6.3	12.3
Seasonal sharing	326	29.6	0.31	-10.3	-0.5	9.2
Stress rule	336	23.2	0.30	-3.3	6.8	16.9
Phenology rule	382	30.1	0.30	-7.5	4.0	15.4
Tuned phenology rule	415	31.6	0.31	-7.0	5.5	17.9
SALS	438	31.6	0.31	-5.6	7.6	20.7
Conventional PV plant	1306	58.7	3.60	16.0	55.2	94.4

Table 17: One-at-a-time sensitivity of the net value (kUSD ha⁻¹ yr⁻¹, relative to the open field) at 90 USD MWh⁻¹, densities designed for the floor as in the main analysis. The capex of the conventional plant is 0.75/0.70 of that of the agrivoltaic plant in the capex rows.

Parameter	Value	Static arrays	Tuned rule	SALS	Conventional plant
Base		6.3	5.5	7.6	55.2
AV capex (USD/Wp)	0.7	9.7	14.9	17.1	72.8
AV capex (USD/Wp)	1.5	0.6	-10.3	-8.2	25.8
Discount rate	4%	8.0	10.3	12.4	64.1
Discount rate	8%	4.4	0.2	2.4	45.4
Crop prices	0.5×	6.4	5.6	7.7	57.0
Crop prices	2.0×	6.0	5.2	7.3	51.6
Electricity price	60	0.3	-7.0	-5.6	16.0
Electricity price	120	12.3	17.9	20.7	94.4

Table A.18: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92